

Do **not** write solutions on this page.

11. [Maximum mark: 18]

The points $A(1, -4, 0)$, $B(-3, -6, 2)$, $C(-1, -2, 4)$ and D form a parallelogram, $ABCD$, where D is diagonally opposite B .

(a) Find the coordinates of D . [2]

The diagonals of the parallelogram, $[AC]$ and $[BD]$, intersect at point E .

(b) Find the coordinates of E . [2]

(c) (i) Given that $\vec{AB} \times \vec{AD} = m \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, where $m \in \mathbb{Z}^+$, find the value of m .

(ii) Hence, find the area of parallelogram $ABCD$. [4]

The plane, Π_1 , contains the parallelogram $ABCD$.

(d) Find the Cartesian equation of Π_1 . [2]

A second plane, Π_2 , has Cartesian equation $5x + y - 7z = 1$.

The acute angle between Π_1 and Π_2 is θ .

(e) Show that $\cos \theta = \frac{1}{5}$. [3]

The line L passes through E and is perpendicular to Π_1 .

The line L intersects the plane Π_2 at point F .

(f) Find the coordinates of F . [5]

